

Instructor: Date:

Learner: Regional Branch:

Program: Center:

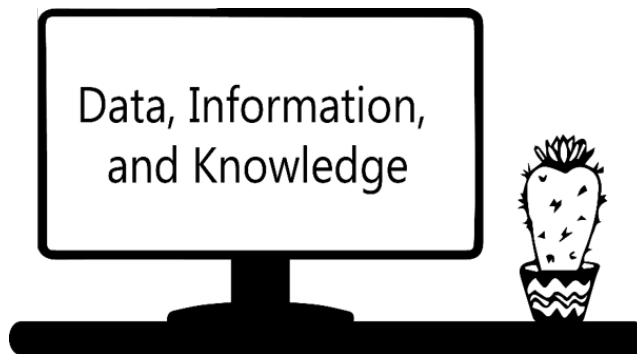
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Let's Start!

Understanding data, information and knowledge.



Learning Outcome:

Describe common activities at work to a using short sentences in familiar spoken and written situations.

Make the result report from the analysis of the information system, based on the customer requirements, following established rules and protocols.

Can Do Statements:

- I can read about data, information and knowledge.
- I can write a short paragraph about data, information and technology.
- I can describe the differences among data, information and technology.

Ask your partner:

- What is the role of a software developer or programmer in information security?
- Do you keep any backup copy of the important files in your computer??





Let's Explore!

Understanding data, information and knowledge.



Instructions: Read the text below.

In today's lesson, we are going to talk about some differences between data, information, and knowledge; they are some of the most important concepts related to technology and the internet.

Data are facts. Every fact that we perceive through our senses is data; data are a reflection and depiction of the world itself. Information is how we capture data, and even though data is real, information can be false sometimes. Let's take a hypothetical situation as an example to explain this matter better.

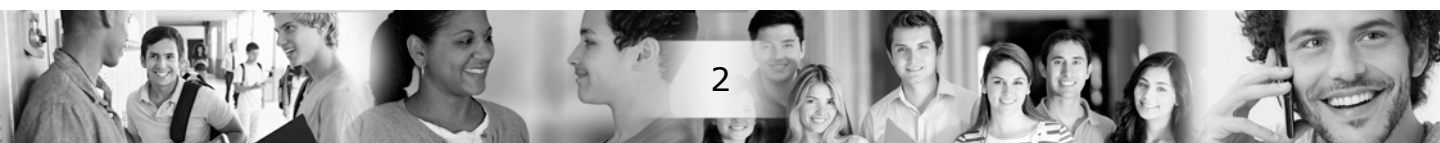
Laura is accessing one of many web pages. There is a basic form that requests Laura's name, last name, age, address and nationality. Every single fact about Laura is data, but the whole set that the web page stores is information.

Let's suppose that someone accesses the server and accidentally erases what Laura typed; this is a loss of information, not of data. Data are still the same; Laura is still Laura Williams, she's 26, she lives at 13 Oak Street, and she is American. However, the information is no longer there; the first difference between data and information is that the information can get lost but not the data.

Let's suppose now that Laura filled the form with the requested data, but she is moving to another country next month. At this very moment, the information is real, but one month from now, it is going to be false. Second difference: the information, as it reflects a single point in time, remains the same but the data change.

Knowledge is a whole set of processes that a brain performs by using data and information. We know the stress load that carries performing many tasks at the same time and expose ourselves to much noise in our workplace. We can know it, maybe because we performed many tasks before and because someone told us their situation related to the exposition to much noise. We complement that information with every aspect related to it that we experience in our daily lives.

Please read the information I'm sending to your e-mails today in the afternoon; we are going to discuss further information next week.



**Let's Practice!**

Reading About Data, Information, and Knowledge.

Instructions: Circle the correct option to complete the sentences.

There are (some / much) differences between data, information and knowledge. Data involves (much / many) real symbols, signs, and facts, meanwhile information comprises (some / any) sets of data. Knowledge has to do with (many / much) brain functions; it includes the decisions you take based on the data you perceive and the information you have.

**Let's Practice!**

Differences Between Data, Information, and Knowledge.

Instructions: Write the words to correctly complete the sentences. Use words from the box below: currently creating, is processing, is going to.

1. One month from now, the data _____ be different from the data.
2. I am _____ information out of pieces of data.
3. Kevin _____ information and data on Monday.





Extension Activity

Differences among concepts.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Read about data, information and knowledge.
- Make a chart with three columns. Use a column for each term. Under each term, write its characteristics.
- Write sentences saying how one concept differs from other.
- Show your chart to your classmates and teacher.

Self-Assessment

Yes

No

- I recognize different concepts.
- I summarize their characteristics.
- I distinguish differences among terms.
- I produce graphic and text to explain concepts.

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Target Words

- **store** (verb): to put things in a special place so they can be used in the future
e.g. Data on one hand refers to the symbols or signs, representing signals, data is not only symbols stored, and data requires objective facts or observations.
- **endowed with** (expression): to have a particular quality or feature
e.g. Information is defined as data that are endowed with meaning and purpose.
- **further** (verb): to a greater distance, or at a more advanced level
e.g. We are going to discuss any further questions in our meeting next week.
- **lack of** (noun): something that is not available or that there is not enough of it
e.g. The data lacks of context and interpretation and is considered simply raw facts.
- **inferred** (noun): to have an opinion or guess that something is true based on the information that you have
e.g. Information is often understood or inferred from data.
- **capture** (verb): to take something into your possession
e.g. Information captures data at a single point while data changes over time.



Learning Points

Countable and Uncountable Nouns.

Countable nouns are the things you can put in an exact amount. Uncountable nouns need a more specific unit of measure.

Some nouns are countable; we can use them in singular and plural forms, and determine them with cardinal numbers.

Example:

A computer – **The** computer(s) – **Many** computers – **Some** computers

An image – **The** image(s) – **Many** images – **Some** images

Some other nouns are uncountable; we cannot determine them by using numbers, and they have no plural form.

Example:

A piece of advice – **The** piece(s) of advice – **Much** advice – **Some** advice

An ounce of prevention – **The** ounce(s) of prevention – **Much** prevention – **Some** prevention

As you could see in the examples, we used articles and quantifiers at the beginning of each clause. Some of them work for both countable and uncountable nouns while some others not. Let's have a brief look at those words:

Definite article **the**: We use **the** definite article the to determine singular and plural.

The report – **The** reports

Indefinite article **a/an**: We use the indefinite article **a/an** to determine a single unit of something; this means, we can only use it with countable nouns. Remember we choose between **a** or **an** depending on the first sound of the noun we determine.

A report – **A** university – **An** opinion – **An** honest person

Quantifier **many**: We use this quantifier to express the existence of more than one thing when that thing is a countable noun.

Many reports – **Many** opinions

Quantifier **much**: We use this quantifier to express the existence of a set or bunch of something that is uncountable.





Learning Points

Much information – **Much** water

Quantifier **some**:

We can use this quantifier to determine both countable and uncountable nouns.

Some computers – **Some** advice

Examples

- *In today's lesson, we are going to talk about some differences between data, information, and knowledge.*
- *Laura is accessing one of many web pages.*
- *We know the stress load that carries performing many tasks at the same time and expose ourselves to much noise in our workplace.*

Be Going to and Present Continuous With Time Expressions.

We use **be going to** to talk about future actions. We can use the present continuous either to talk about future or present actions or plans. We can determine the time by using certain time expressions.

We use **be going to** to express a future idea or plan when we intend to do something or have decided to do something but did not arrange it. When we use this structure, we focus on the intention of performing the activity.

Subject + **verb to be** + **going to** + complement

We use the **present continuous** when we say what we have planned and arranged to do at a specific time in the future. When we use this structure, we focus on the **arrangement** of the activity.

Subject + **verb to be** + **verb -ing** + complement

We can use certain **time expressions** with both of these structures.

Time expressions with the connotation of **present** tense.

Now, right now, at present, currently, at the moment, nowadays, this year.

Time expressions with the connotation of **future** tense.

Tonight, tomorrow, today, on (Monday – Sunday), in (January – December) tonight, next week, next year.

Examples:

Now, I'm going to explain to you the difference between data and information.

You **are presenting** this topic **on Wednesday**.

At the moment, I **am reading** an article on the difference between data, information and knowledge.

